

Aruna: Improving fishermen's livelihoods with global market access and data insights

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Aruna grows its digital fishing ecosystem on Google Cloud by connecting fishermen with buyers in global markets, and promotes sustainable fishing by matching supply with demand based on data insights.

About Aruna

Founded in 2016, Aruna is the first start-up to connect Indonesian fishermen to the global market through its digital fishing ecosystem. By promoting fair trade and sustainable fishing, the company seeks to improve the livelihoods of fishermen and their families. As part of its holistic approach to community empowerment, Aruna also employs women from fishing communities to support supply point operations. The company has won numerous awards for revolutionizing the fishing industry, including the Asean ICT Awards for Private Sector in 2018, and

Google Cloud results

- Digitizes fishing processes to connect more than 36,000 fishermen to global markets
- Improves fishermen's knowledge of sustainable fishing by visualizing protected marine areas and vessel movements with Google Earth Engine
- Reduces food waste by predicting demand for thousands

10X in fishery revenue data

more recently, Fast Company's Ten most innovative Asia-Pacific companies for 2022.

Industries: Retail & Consumer Goods

Location: Indonesia

(<https://workspace.google.com/products/drive/>)

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

~~Google workspace~~

Google Drive

(<https://workspace.google.com/products/drive/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace

(<https://workspace.google.com/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

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Google Earth Engine

(<https://earthengine.google.com/>)

of SKUs with BigQuery

In Indonesia, fishermen rely on traditional knowledge to read the weather and locate prime fishing grounds. How much is caught and sold is left to chance and at the mercy of elements, with increasing temperatures and unseasonal weather patterns caused by climate change further impacting fishermen's catches.

Aruna (<https://aruna.id/>), a fishery products supply chain, offers data and tools to help fishermen navigate uncertainty and maximize their revenue. The company connects more than 36,000 Indonesian fishermen with global markets, and gives them insights into sustainable fishing through its applications, Aruna Heroes and Nelayan, on **Google Cloud** (<https://cloud.google.com/>).

With the Aruna Heroes app, local tech-champions help fishermen log catches, record fish sales, bid for the best price, and receive prompt payments. The app allows users to enter transactions even when there's no internet connection. As soon as the app is back online, data is automatically synced with Google Cloud to ensure consistency across the Aruna ecosystem.

The Nelayan Aruna app is where local eco-champions provide fishermen with information on sustainable fishing methods and regulations. In 2021, Aruna launched GPS trackers to help

fishermen avoid illegal fishing and improve their profit with seafood traceability.

"With Google Cloud, we can handle all of our computing and data needs in one place, allowing us to focus on building an ecosystem that improves fishermen's livelihoods," says Walesa Danto, Head of Products at Aruna.

He adds, "Our frontend apps and business systems can scale quickly without downtime using [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine)

(<https://cloud.google.com/kubernetes-engine>). With [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>), we can help fishermen make better decisions about where and what to fish, and improve sustainable fishing."

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Transforming the fishing industry with digital literacy

Traditionally, fishermen in remote villages sold their catch to middlemen who would then store and transport the fish to a larger market. Through Aruna, fishermen gain direct access to wholesalers and consumers throughout Indonesia and abroad, without worrying about logistics and export clearance. According to the Center for Impact Investing and Practices, fishermen increase their income by three to twelve times

(https://ciip.com.sg/docs/default-source/default-document-library/scaling-impact-in-asia-2022-case-study-aruna-ciip-accenture-smu.pdf?sfvrsn=474e6ed4_4)

after joining Aruna.

However, in its efforts to transform traditional fishing with a mobile app, Aruna faced several challenges. Only 25% of fishermen use smartphones in rural villages due to limited power supplies and low digital literacy. Besides working with local governments to improve technology access with solar panels and satellite internet, Aruna recruited mobile-savvy youths to help fishermen go digital.

To date, Aruna has helped more than 36,000 fishermen in 150 spots across 27 provinces in Indonesia gain access to global markets.

Scaling a seafood supply chain for global opportunities

Paper-based transactions in the fishing industry historically slowed the flow of information, making it challenging to trace seafood from the fishermen to the consumer. It is difficult for fishermen to export fish abroad if they do not understand the paperwork or logistics involved.

By digitizing processes, Aruna facilitates collaboration among hundreds of sites in the fishery ecosystem. For example, catch data from fishermen on the Aruna Heroes app can be used to secure required documents for export clearance.

Boosting sales and efficiency also requires a seamless flow of information across Aruna's supply chain, from suppliers to logistics providers, and from warehouses to customs. For example, if the Warehouse Management System is down, the ecommerce website cannot access stock data in real time, which prevents consumers from completing their transactions.

"In order for our business to operate smoothly, every system must function properly," says Danto. "Google Kubernetes Engine (GKE) ensures that every system has a backup so we can run our operations smoothly without interruption. When we experience CPU spikes from concurrent queries or heavy

workloads, GKE automatically deploys a container in minutes to meet service-level agreements (SLAs) with partners."

Aruna runs more than 200 microservices on GKE, including a message bus service to facilitate communication between apps and the Know-Your-Customer (KYC) process to onboard Aruna Fishers.

A One Google approach for data insights to maximize sales

Besides setting up a digital platform, Aruna standardized stock-keeping units (SKUs) so everyone reports information in the same way across the supply chain. Each SKU represents a product with specific criteria such as fish species, size, and packaging type.

The company gains supply chain visibility by tracking more than 1,000 SKUs in its Odoo Enterprise Resource Planning (ERP) system, from the moment the fish arrives at the dock right up to when the packaged product leaves the warehouse. Using BigQuery, Aruna collates real-time information from Odoo and other business systems, and secures data with role-based access to grant users the right amount of access. By accurately predicting demand, the company can maintain the correct stock level for each SKU and reduce food waste.

"Tight integration between Google Cloud products helps us to query and process large amounts of raw data for quick insights," says Danto. "We extract, transform, and load (ETL) more than 200 Google

Sheets

(https://workspace.google.com/intl/en_sg/products/sheets/)

worth of data stored on Google Drive

(https://workspace.google.com/intl/en_sg/products/drive/)

, to BigQuery in less than 30 minutes." He goes on to add that with Aruna's current data infrastructure, data can be easily managed with "drag-and-drop" in Google Drive.

Aruna's predictive analytics model involves data from various sources, including historical transactions in Google Sheets, inventory data from its warehouse system, weather data from Google Earth Engine, and location data from GPS trackers. Even for cases where Aruna Heroes record catch data offline because of power shortages, BigQuery can simulate the lagging data before the catch arrives at the stage in the supply chain.

For instance, a retailer wants to buy 200 lobsters in three days, but the current inventory only shows 150 lobsters are available. By forecasting inventory, Aruna knows whether it has enough lobsters to fulfill the customer's order by the date specified.

As part of its digital marketing strategy, Aruna also utilizes Google Ads (<https://ads.google.com/home/>) and YouTube (<https://www.youtube.com/>) to create brand awareness and reach new customers. By analyzing performance data such as conversion tracking and audience engagement, Aruna ensures that its ads and content are targeted at the right audience.

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Ensuring fishing sustainability with earth-scale analysis

Fishing sustainably is a crucial use case at Aruna. By 2030, the Ministry of Marine Affairs and Fisheries aims to protect 32.5 million hectares of marine area in Indonesia from overfishing and habitat loss. Aruna helps fishermen stay on top of regulation changes and fish more sustainably.

In a pilot program, more than 300 fishermen share their movement data with Aruna through GPS trackers installed on their fishing boats. Aruna imports the GPS data into Google Earth Engine (<https://earthengine.google.com/>) to visualize vessel movement with satellite imagery of Indonesia's maritime territory and weather.

Aruna uses the location data to verify that fishermen are fishing within permitted areas when reporting seafood sustainability to regulators and global certification bodies. Additionally, fishermen can meet the increasing demand for sustainably sourced seafood by providing information about the catch's time, date, and location.

As more fishermen share their data through the program, Aruna aims to apply machine learning algorithms to examine how variables such as weather and fishing effort can affect fishermen's earnings. Using data insights, Aruna recommends ways for fishermen to improve their performance.

"We are on the right path, especially in terms of technology, as we expand our ecosystem and create new services," says Utari Octavianty, General Director and Co-Founder at Aruna. "With Google Cloud, our apps and systems run faster, and we can track

omnichannel activities simultaneously via one platform."

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